

CSE-364 (Software Development Project-1)

Group A2 (ASMR)

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1. Project Name

Campus Hub: A platform to automate academic tasks, including course advising, attendance updates, scholarship processing, and AI-based student support.

2. Introduction

The use of technology at educational institutions has brought about immense changes in the way each educational institution manages its work and teaching worldwide. Yet, in Bangladesh, most universities still rely on an older manual system or different software to independently maintain academics. These systems often need to meet the ever-growing requirements of students and teachers; thus, problems arise in accessing important ser- vices.

The automation of course advisement itself is crucial, as students can get recommendations based on their academic performance, thus helping to improve and reduce dropout rates. According to the University Grants Commission of Bangladesh, in a 2022 survey, more than 60% of colleges and universities have various problems regarding student services, such as course registration, attendance, and academic advising, due to their obsolete systems (UGC, 2022)[1]. Integrating leave applications with auto-attendance remark systems helps simplify administrative tasks and brings more transparency into view.

Course registration is a significant academic activity; however, many systems generally lack user-friendly interfaces and student personalization.

A study by the Bangladesh Bureau of Educational Information and Statistics (BAN- BEIS) showed that more than 70% of university students in Bangladesh have trouble getting timely help with their studies, scholarship applications, and clear information about exams and notices (BANBEIS, 2022)[2]. These problems often cause missed deadlines, lower academic performance, and unhappiness with educational experiences. The application processes for scholarship facilities overwhelm students and drain time for the administrators. An online form submission system streamlines this process, promoting equity and accessibility. Exam scheduling and notifications often need clarification; a dynamic, editable calendar provides clarity and ensures timely updates. Likewise, an efficient digital notice board allows for direct communication among administration, faculty, and students.

A report by the World Bank on higher education in Bangladesh identified that the use of AI and automation has been particularly instrumental in easing the process of university administration (World Bank, 2021)[3]. Without AI-powered support systems, students have had difficulty accessing swift and accurate information related to academic regulations, course selection, and administrative procedures. AI-powered chat support incorporated into the proposed system greatly improves the user experience by availing university-related information

3. Related Project

1. PeopleSoft Campus Solutions:

PeopleSoft Campus Solutions by Oracle is a robust Student Information System (SIS) commonly used in higher education institutions. It combines various features to simplify student administration processes.

- **Automated Course Advisor:** Evaluates student performance, highlights failed courses for retakes, and pinpoints areas for GPA enhancement.
- **Course Registration:** Students can easily select, drop, or modify courses for each term through a user-friendly online portal.
- **Leave Application:** Streamlines leave requests, with approvals automatically updated in the attendance system.
- **Exam Scheduler:** Offers a calendar view of scheduled exams, which can be modified by teachers or administrators.
- **Notice Board:** Administrators and faculty can post announcements directly to student dashboards.
- **AI Chat Support:** Some implementations include AI chatbots to respond to common student inquiries regarding courses, registration, and schedules.

2. Ellucian Banner:

Ellucian Banner is a comprehensive ERP system tailored for higher education, aimed at streamlining student services and administrative functions.

- **Automated Course Advisor:** Provides tools for academic advising, tracks student progress, and suggests courses for retaking based on GPA and previous failures.
- **Course Registration**: Students can easily select courses, register for upcoming terms, and check their class schedules.
- **Leave Application**: Allows students to submit leave requests, with automated updates to attendance records once approved.
- **Scholarship Application**: Includes integrated forms that enable students to apply for scholarships and monitor the status of their applications.
- **Exam Scheduler**: A built-in calendar tool shows exam schedules, which can be updated by teachers and administrators.
- **Notice Board**: Administrators can send notices directly to students' portals.
- **AI Chat Support**: Enhanced with AI chatbots to assist with common inquiries.

3. Blackboard Learn:

Blackboard Learn is a widely used Learning Management System (LMS) offering various tools focused on enhancing the student experience for both academic and administrative tasks.

- **Course Registration:** Allows students to easily enroll in or withdraw from courses through a self-service portal.
- **Automated Course Advisor:** Connects with academic advising tools to recommend courses based on students' grades and GPA.

- **Leave Application:** Includes plugins for managing leave requests, which facilitate approval workflows and track attendance.
- **Exam Scheduler:** Teachers can update exam schedules, which students can view through integrated calendar features.
- **Notice Board:** Enables teachers and administrators to post important course notices or university announcements.
- **Scholarship Application:** The platform can incorporate forms for scholarship applications, although the specific functionality may vary based on customization.
- **AI Chat Support:** This offers support for virtual assistants or third-party integrations, such as IBM Watson, to help answer student FAQ.

4. Moodle LMS:

Moodle is a widely-used open-source Learning Management System designed for academic settings in universities around the globe.

- **Automated Course Advisor:** Custom plugins can evaluate student grades to recommend courses for retakes or options for improving GPA.
- **Course Registration:** Through Moodle's enrollment modules, students have the ability to add, drop, or adjust their course selections.
- **Leave Application**: Custom leave management plugins enable students to submit leave requests, which automatically update attendance records once approved.
- **Exam Scheduler**: Includes a calendar that allows teachers to update and publish exam dates.
- **Scholarship Application**: Moodle can integrate forms for students to submit their scholarship applications.
- **Notice Board**: Includes forums and announcements for notices from administrators and teachers.
- **AI Chat Support**: Allows for the integration of AI chatbots to assist with student FAQs and administrative inquiries.
- **Leave Application**: Custom leave management plugins enable students to submit leave requests, which automatically update attendance records once approved.
- **Exam Scheduler**: Includes a calendar that allows teachers to update and publish exam dates.
- **Scholarship Application**: Moodle can integrate forms for students to submit their scholarship applications.
- **Notice Board**: Includes forums and announcements for notices from administrators and teachers.

- **AI Chat Support**: Allows for the integration of AI chatbots to assist with student FAQs and administrative inquiries.

4. Objectives

- Automate course advising by analyzing student results and suggesting courses for retaking or improvement.
- To automate the leave application process where students can apply for leave, teachers can approve it, and attendance sheets are automatically updated.
- To facilitate course registration by allowing students to select, deselect, and submit their desired courses for each term.
- To provide a scholarship application system where students can fill out and submit scholarship forms easily.
- To offer a centralized exam scheduler with a calendar view displaying all exam schedules, updated by teachers and visible to students.
- To enable seamless communication through a digital notice board where administrators and teachers can broadcast important updates to students.
- To enhance student support services using AI-powered chat support for answering frequently asked questions related to academics and university processes.

5. Features

Automated Course Advisor: Students can input their results, and the system will suggest courses to retake, supplement, or improvement so that students can improve their result.
Leave Application and Auto Remark: Students can apply for leave, and Admin can approve requests. Attendance sheets automatically reflect approved leaves with remarks. Students can see their attendance status.
Course Registration: Students can select or deselect courses and submit a finalized list for the term.
Scholarship Application: Students can fill out and submit scholarship applications seamlessly.
Exam Scheduler: Displays a calendar view of exams. Teachers can edit schedules, while students can only view them.
Notice Board: Administrators and teachers can broadcast notices directly to students.
AI Chat Support: Provides instant responses to queries related to academics or university processes, functioning like a FAQ assistant.

6. Stakeholders/Users

a. Admin
b. Teacher
c. Student

7. Project Architecture Diagram

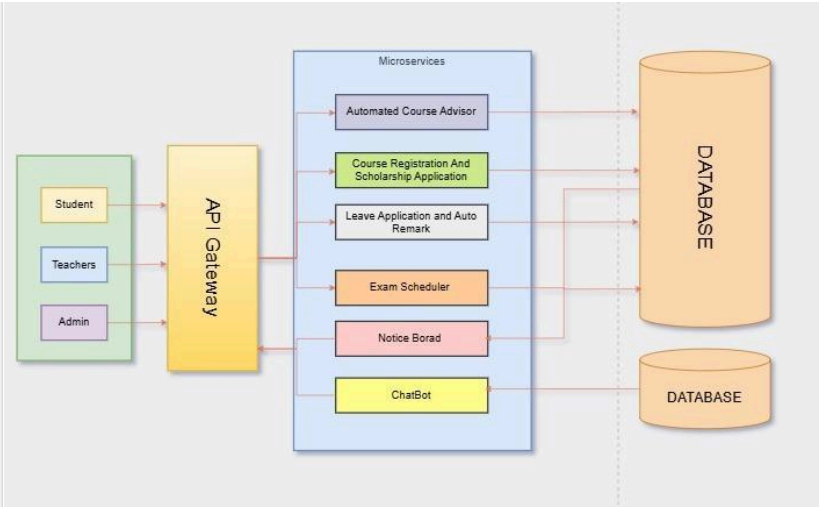


Fig 1: Architecture Diagram

8. Frontend

- The front-end development will utilize modern web technologies, focusing on a responsive and intuitive user interface (UI). We've opted for Next.js.
  - **Framework:** NextJS
  - **Design Tools:** Tailwind CSS
  - **Details:**
    Tailwind CSS: Tailwind CSS is a utility-first CSS framework that streamlines the styling process by providing a set of pre-defined utility classes. It differs from traditional CSS frameworks by focusing on low-level utility classes, allowing developers to rapidly build designs without writing custom CSS. Tailwind CSS is highly customizable and scalable, making it a popular choice for projects like TripFlair where a flexible and utility-driven approach to styling is advantageous.

    Next.js: Next.js is a React framework that simplifies the development of web applications, offering server-side rendering and efficient routing. It enables the creation of dynamic and performant websites by providing a structured framework for building React applications. With features like automatic code splitting and server-side rendering, Next.js optimizes web performance and enhances the overall development experience.

9. Backend

- The backend will utilize Node.js with Express.js for its flexibility and scalability. MongoDB will serve as the database for efficient data storage and retrieval. Real-time communication is facilitated using Socket.io.
  - **Programming Languages:** JavaScript
  - **Database:** PostGreSQL
  - **Frameworks:** ExpressJS, Django

- **Runtime Tool:** Node.js
- **Additional Tools:** Socket.io, NLTK, spaCy, ChatterBot, and Ollama

10. Prototype of the Project

During the prototyping phase of our project, we chose Figma as our design tool due to its numerous advantages:

1. **Seamless Collaboration** – Figma supports real-time collaboration, enabling team members to work simultaneously on designs. This feature is particularly beneficial for our project, allowing multiple stakeholders to refine user interfaces and experiences collectively.

2. **Streamlined Workflow with Components** – By utilizing Figma's component system, we were able to enhance efficiency in our prototyping process, minimize development time, and maintain consistency across different elements.

3. **Interactive and Realistic Prototyping** – Figma's robust prototyping features allowed us to create dynamic user flows with realistic interactions, making usability testing more effective and insightful.

4. **Version Control for Smooth Development** – The platform's version control capabilities ensure that all design changes are tracked and allow easy rollback to previous versions. This is essential for managing updates in our project, which includes multiple profiles, dashboards, and scheduling functionalities.

5. **Cross-Platform Accessibility** – As a cloud-based tool, Figma allows team members to access and edit designs from any device, ensuring flexibility and uninterrupted workflow, regardless of location.

11. Cost Estimation

Stage	Estimated Monthly Cost (BDT)	Approximate Time	Total Cost (BDT)
Planning	50,000	1 month	50,000
UI/UX Design	50,000	1 month	50,000
Prototyping	60,000	1 month	60,000
Frontend Development	60,000	3 months	180,000
Backend Development	70,000	3 months	210,000
API Integration	80,000	2 months	160,000
Database	60,000	2 months	120,000
Testing & Quality Assurance	60,000	1 month	60,000
Documentation & Training	40,000	-	40,000
Operation & Maintenance	100,000	-	100,000
Total Cost			**1,030,000BDT**

Table: Cost Estimation

12. Final UI of the Project

The final UI of our project is given below.

This is the selecting role page of our project where students, teachers, and admin can separately log in according to their role.

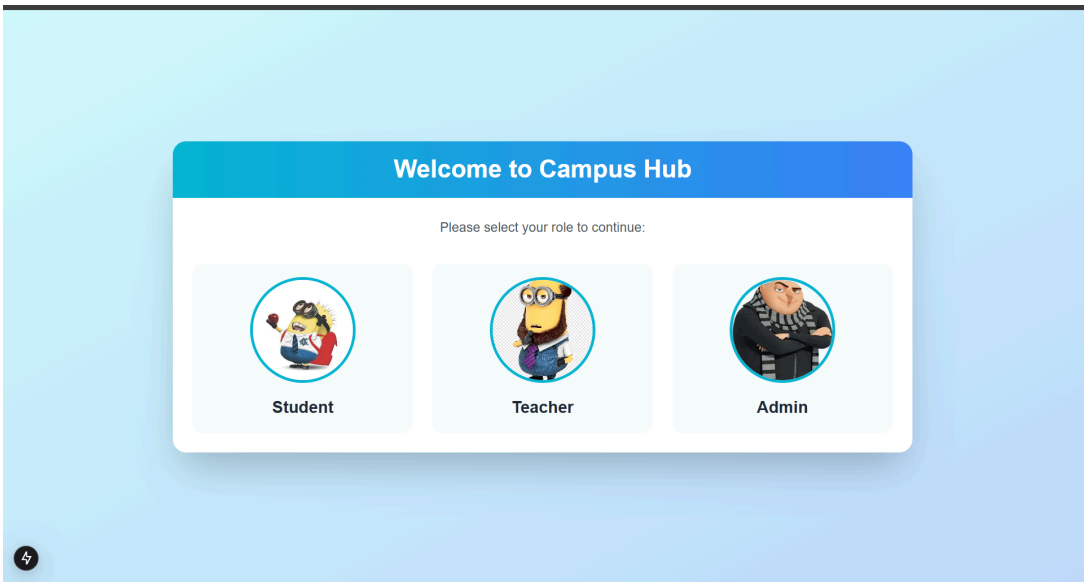


Fig 2: Role Selection Page

After selecting the role, for example, a student has to enter, and then a login page will come

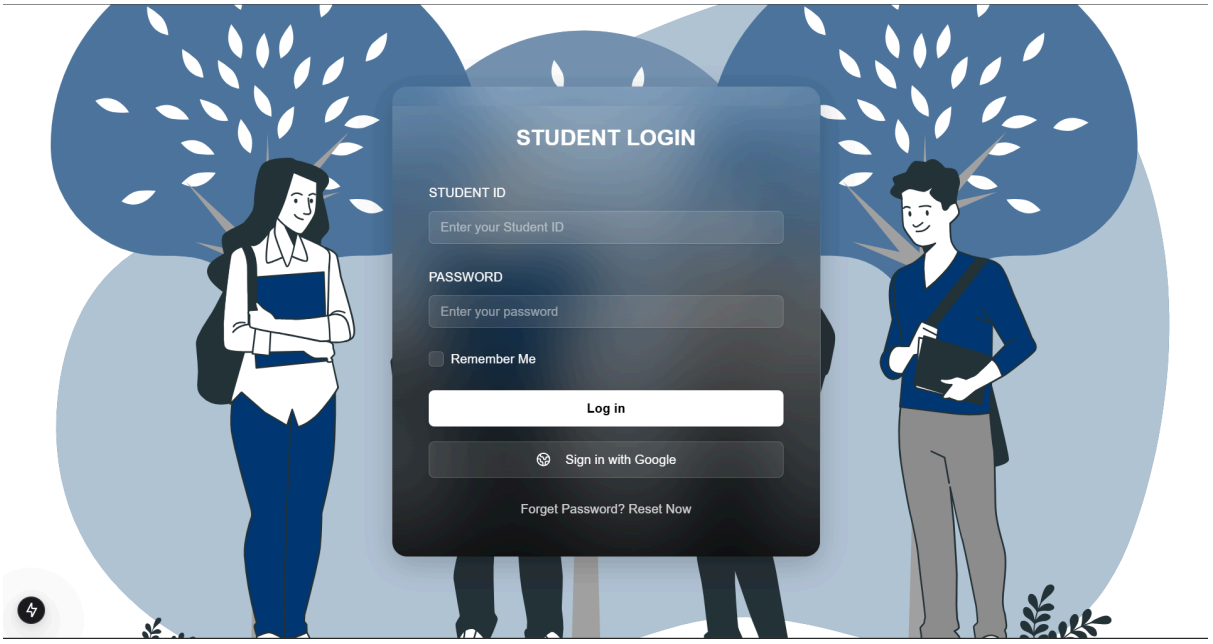


Fig 3: Log In page

After successfully entering the ID and password, the landing page of our project will be shown. Login process is same for teacher and admin as well.

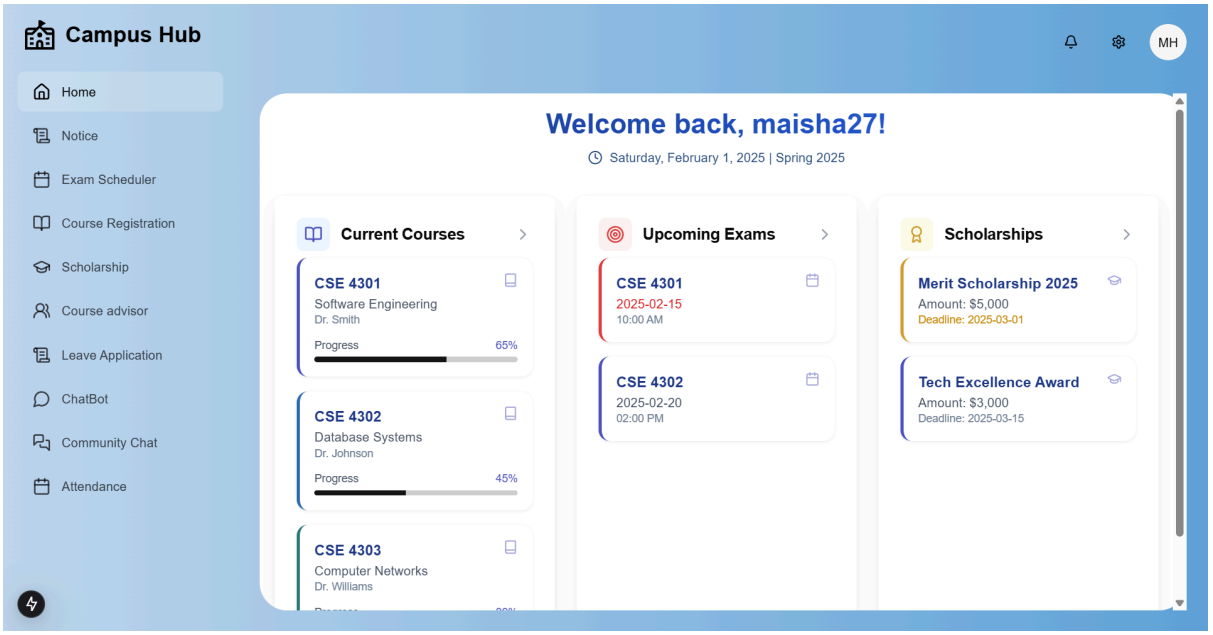


Fig 4: Landing Page

A student can only view the notices, but teachers and admins can add, delete, and update notices.

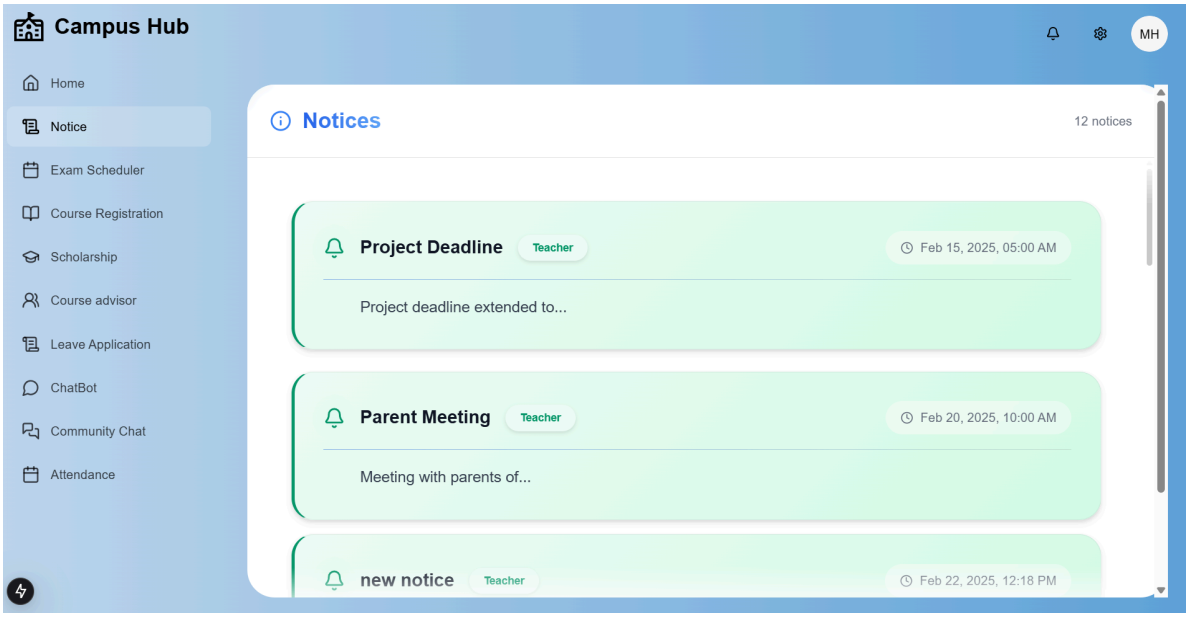


Fig 5: Notice Page (Student)

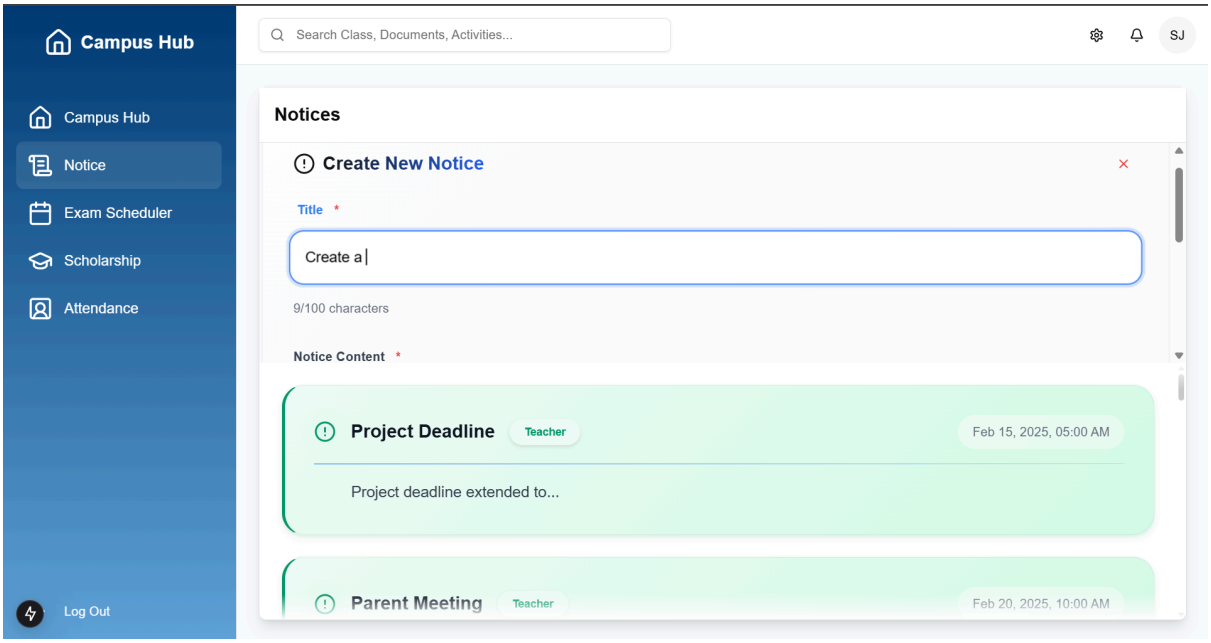


Fig 6: Notice Page (Teacher, Admin)

This is the Exam Scheduler page for teachers, where Teachers schedule exams for students including date, start-time, end-time, and room-number info for students to view on their dashboard. All exams are scheduled within an 8 AM to 3 PM time constraint and Sunday to Thursday weekly constraint. Consistency and clashing are also maintained by handling two teachers who can not schedule exams in the same room at the same time and other corner cases.

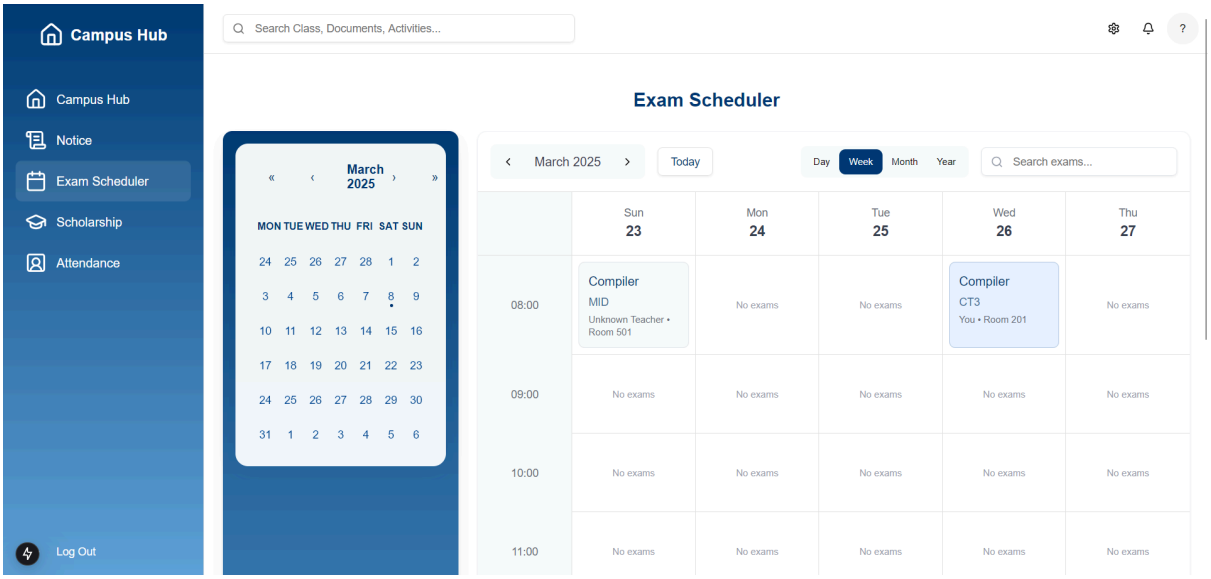


Fig 7: Exam Scheduler Page (Teacher)

This is the exam scheduler page for students. Students can only view scheduled exams, not anything else, since teachers can edit.

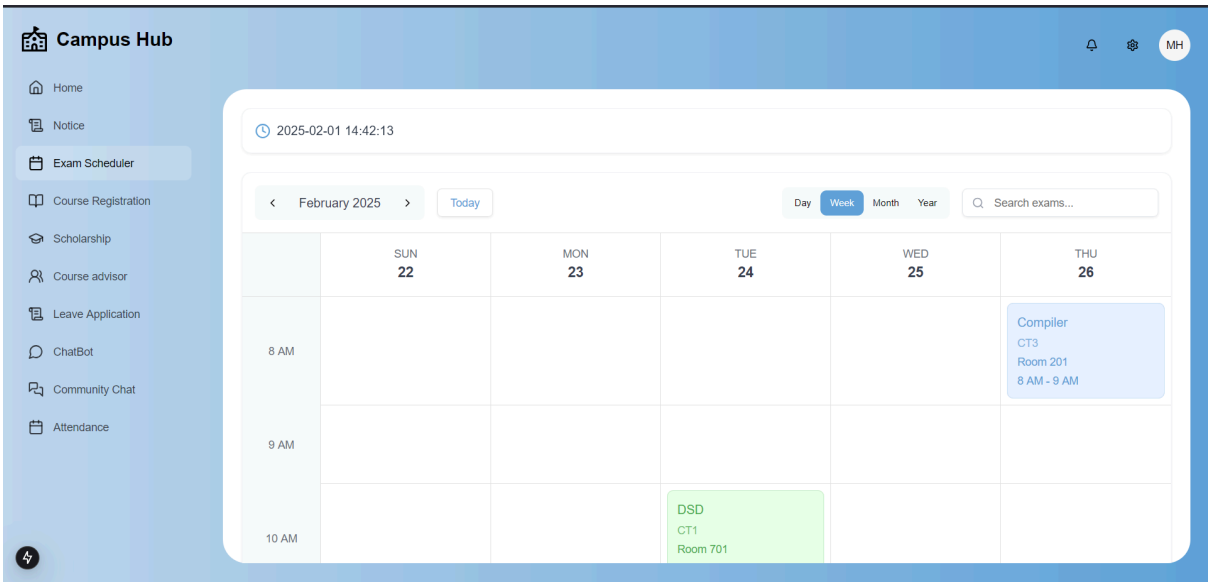


Fig 7: Exam Scheduler Page (Student)

Course Registration page where students can register for courses.

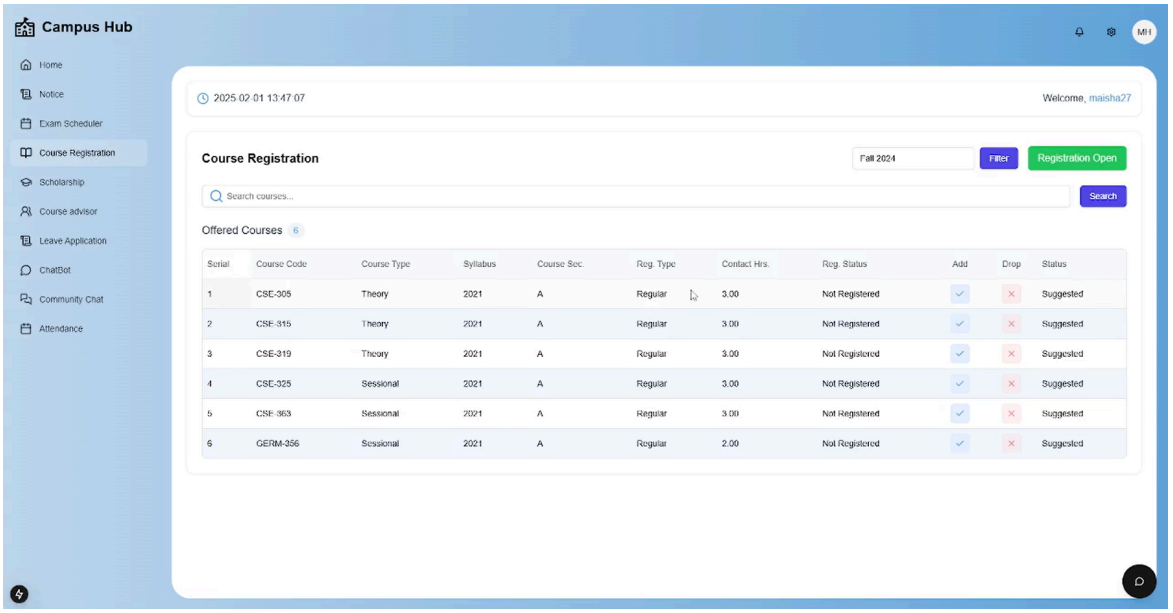


Fig 8: Course Registration

If a student wants to add a course, then he can simply add the course. Search functionality is also provided to enhance the UI.

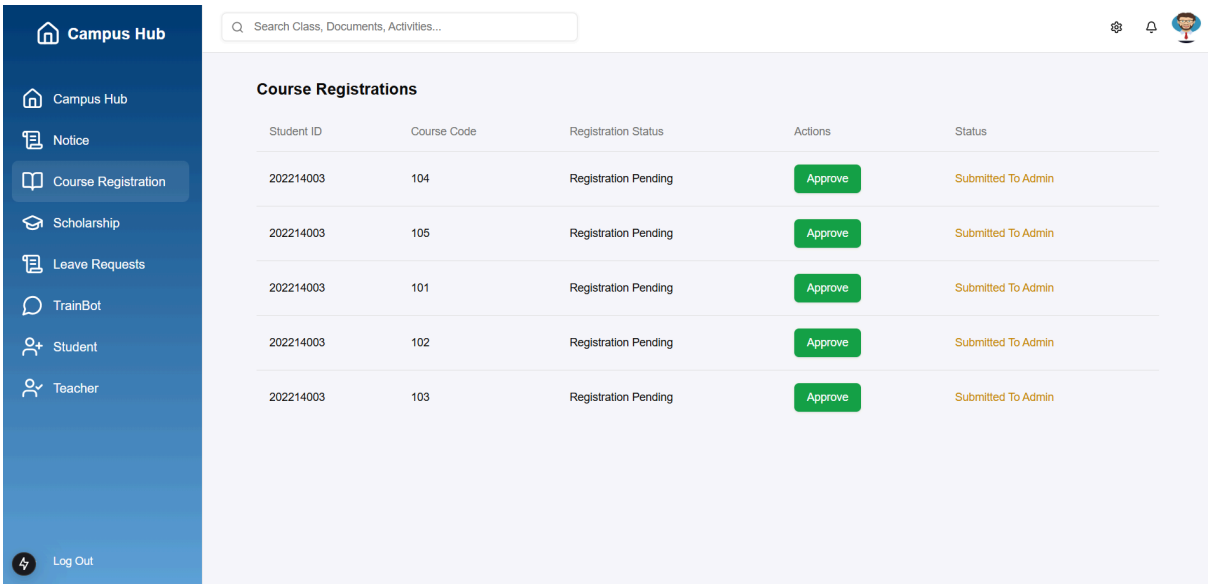


Fig 9: Course Registration View for Admin

After adding or dropping a course, it will be approved by the admin.

Scholarship application is also provided in our project where students can apply for a scholarship after completing the scholarship form, which then will be verified or rejected by the teacher. And final approval from the admin.

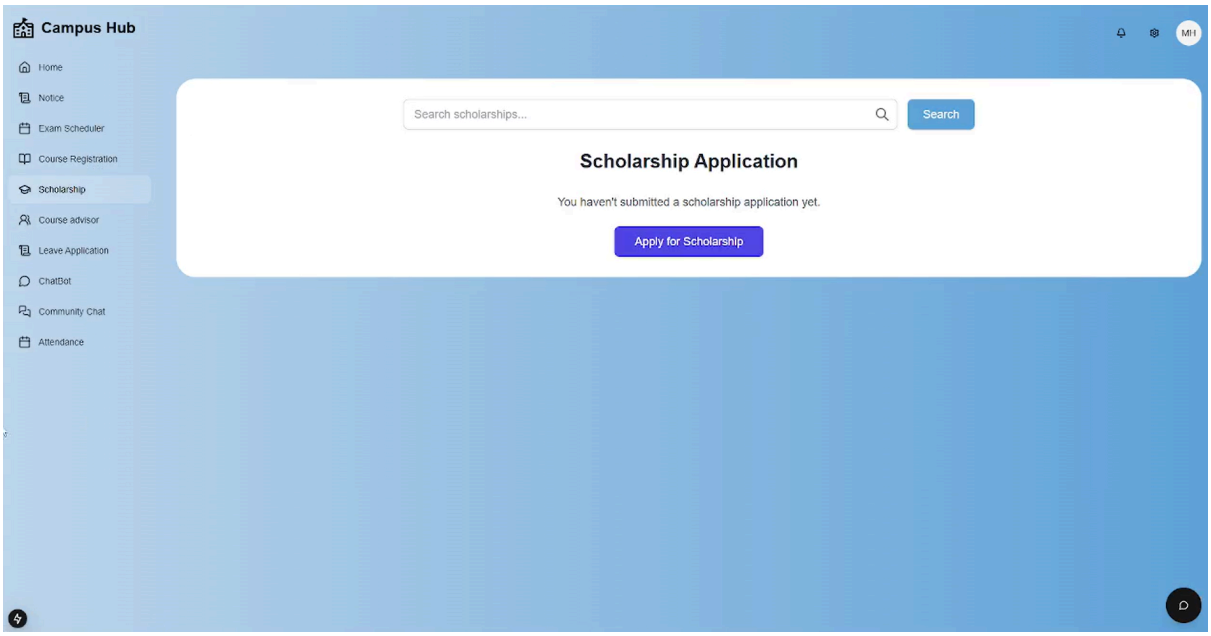


Fig 10(a): Scholarship Application Page

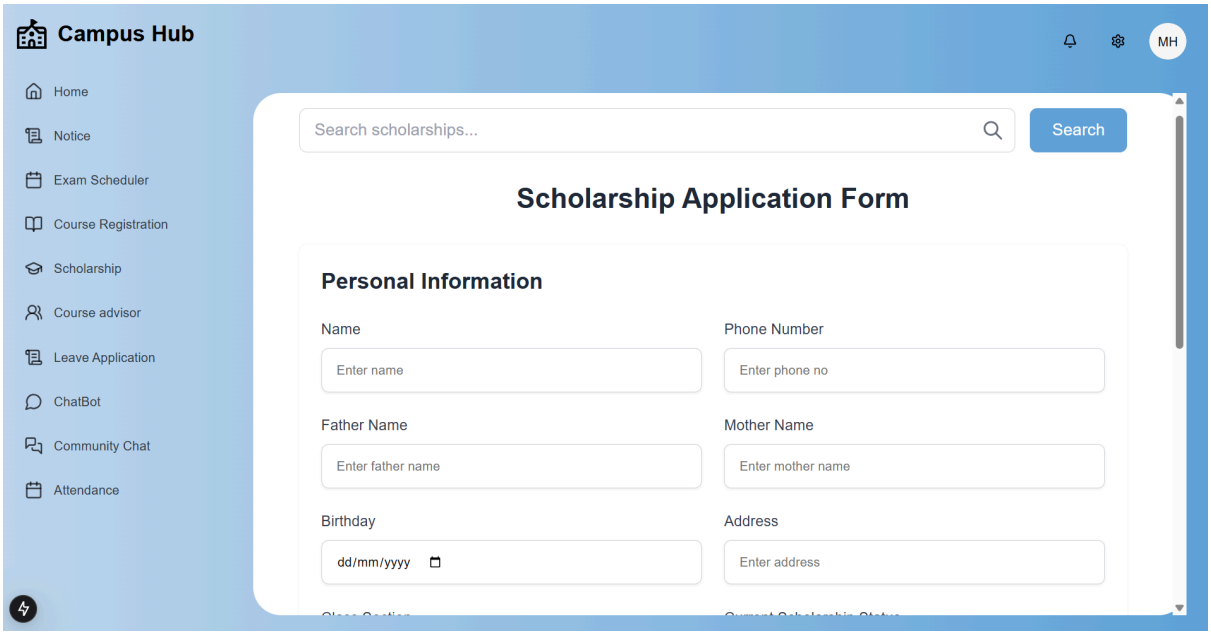


Fig 10(b): Scholarship Application Form

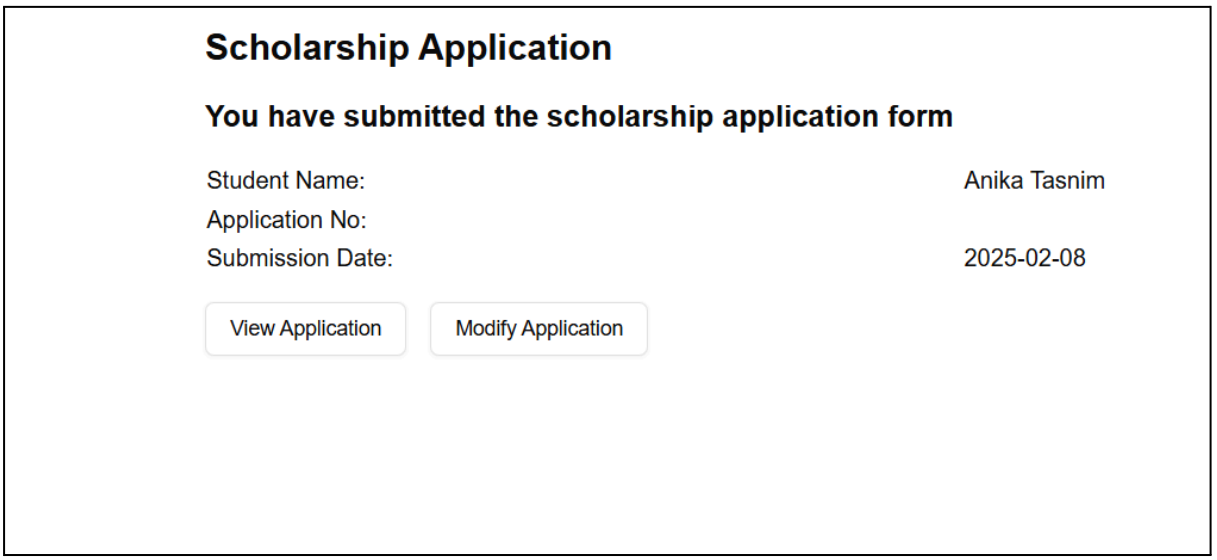


Fig 10(c): Scholarship Application After Submission

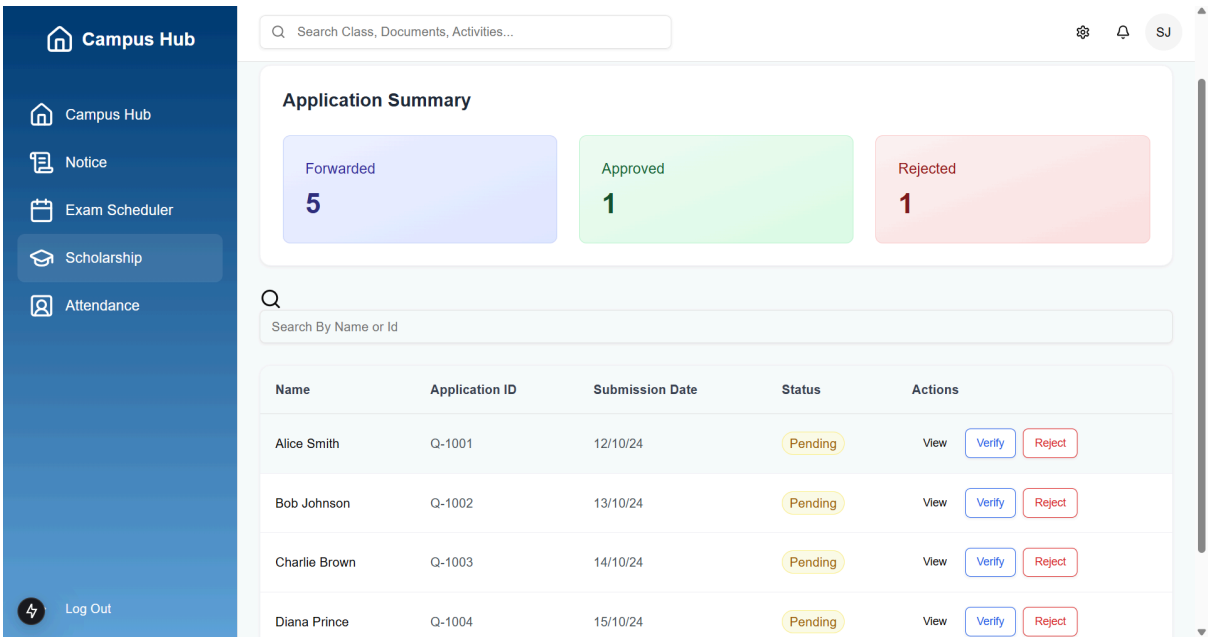


Fig 11: Scholarship View For Teacher

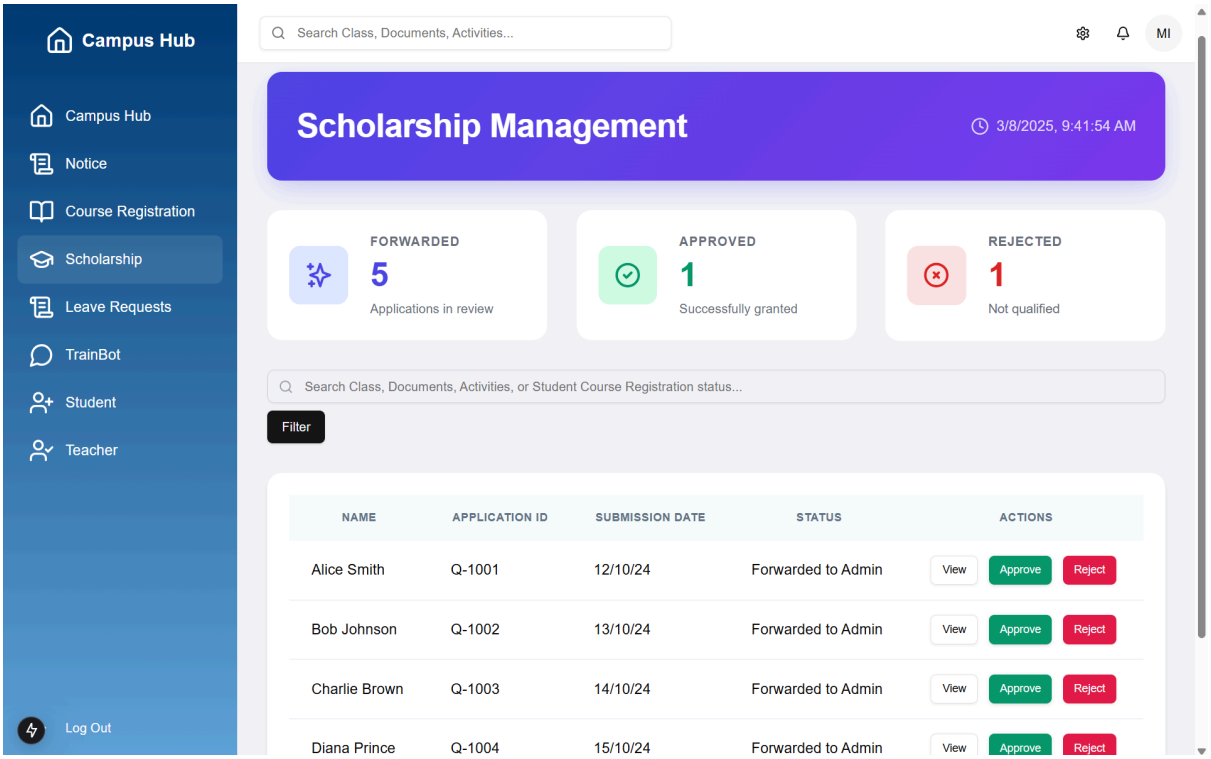


Fig 12: Scholarship View for Admin

Students can also apply for leave by filling out the Leave Application Form. Students can only apply for the course in which he/she is admitted. Student can not submit a leave application which is not a valid class for him/her.

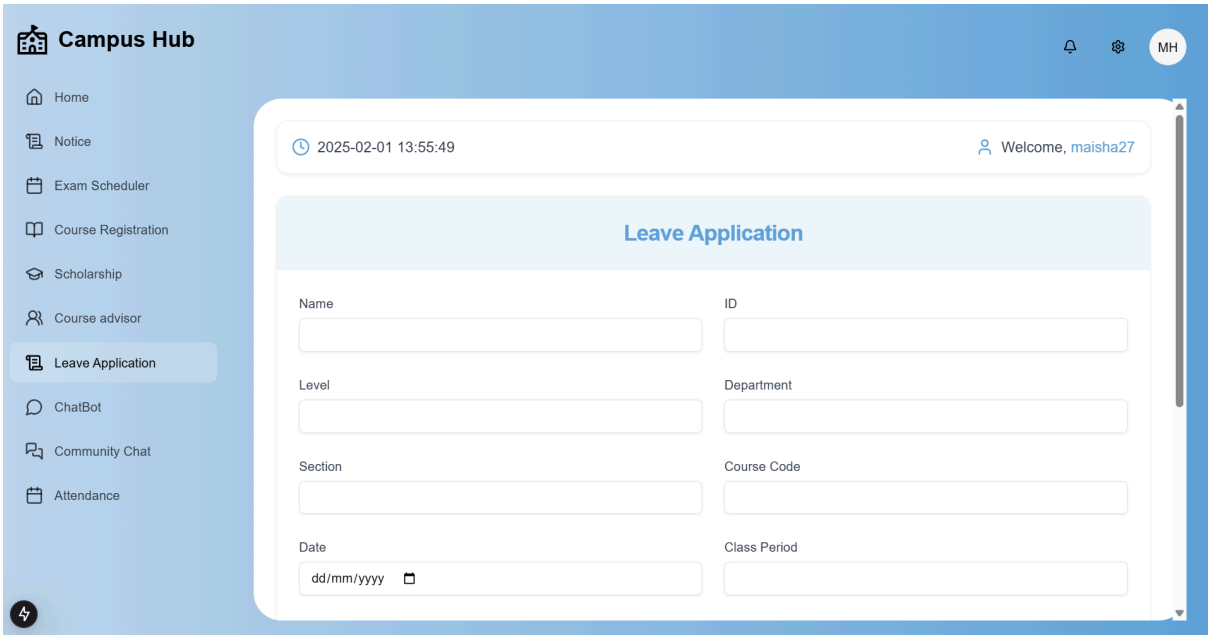


Fig 13: Leave Application Form

This leave application form will be sent to the Admin. Here admin can see all the leave requests and their details.

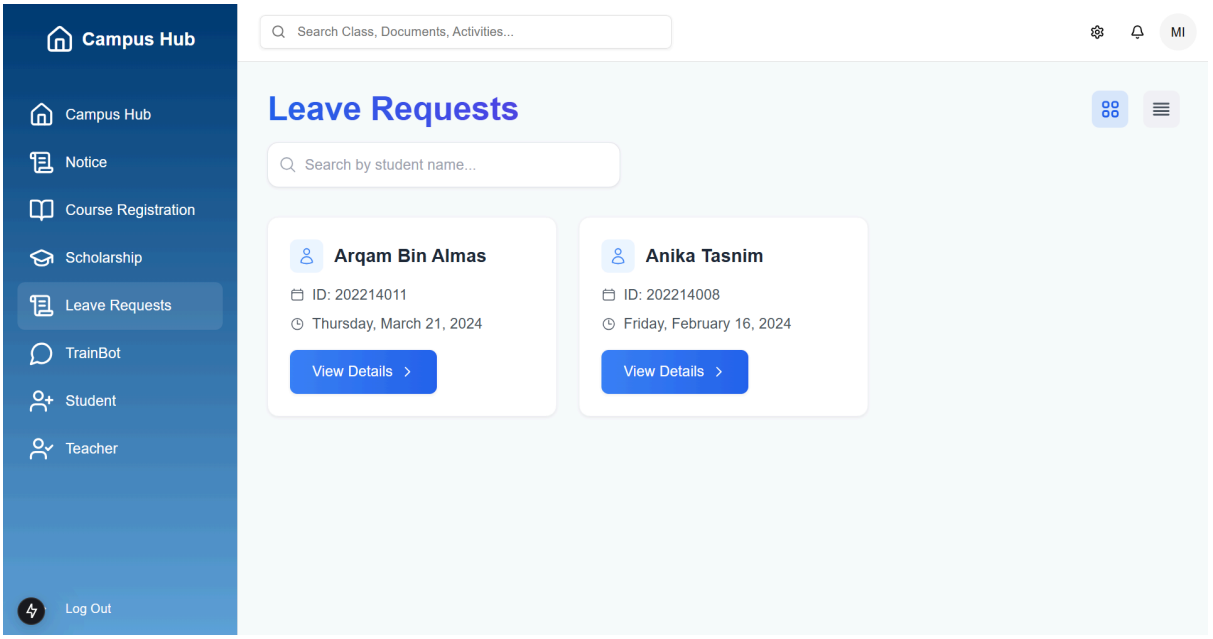


Fig 14: Leave Request list in admin view

Here admin can see the details and based on that, can approve or reject the leave request. And the approval/ rejection will be updated to the attendance of that student.

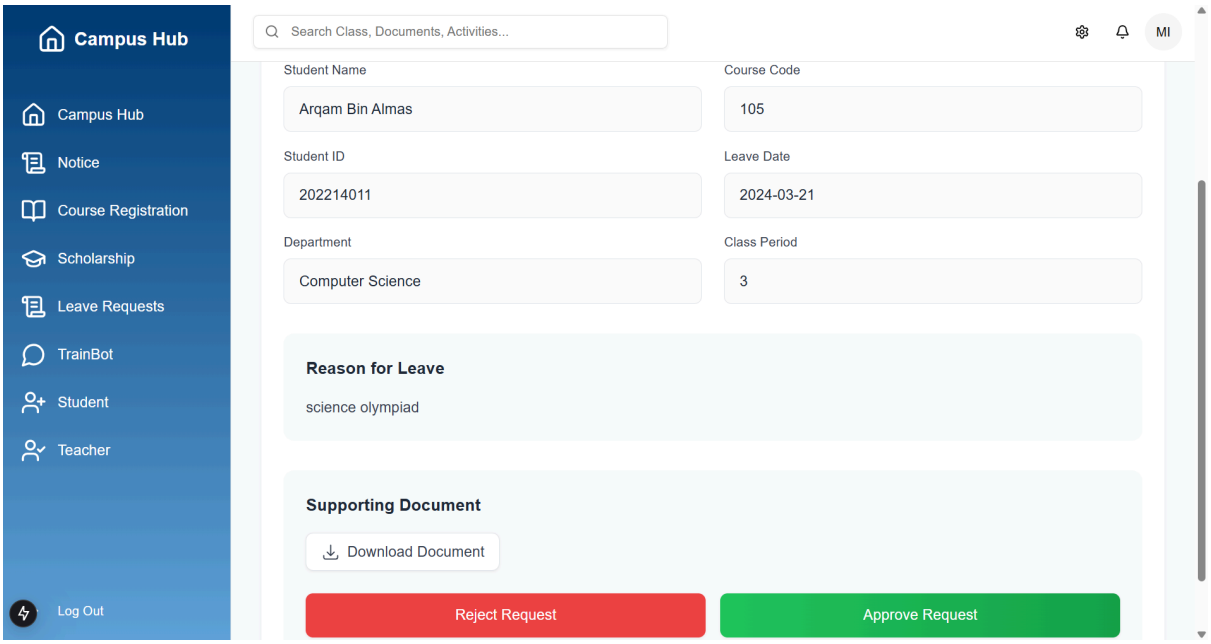


Fig 15: Leave details with approval and rejection functionality

Here students can see attendance of each current course.

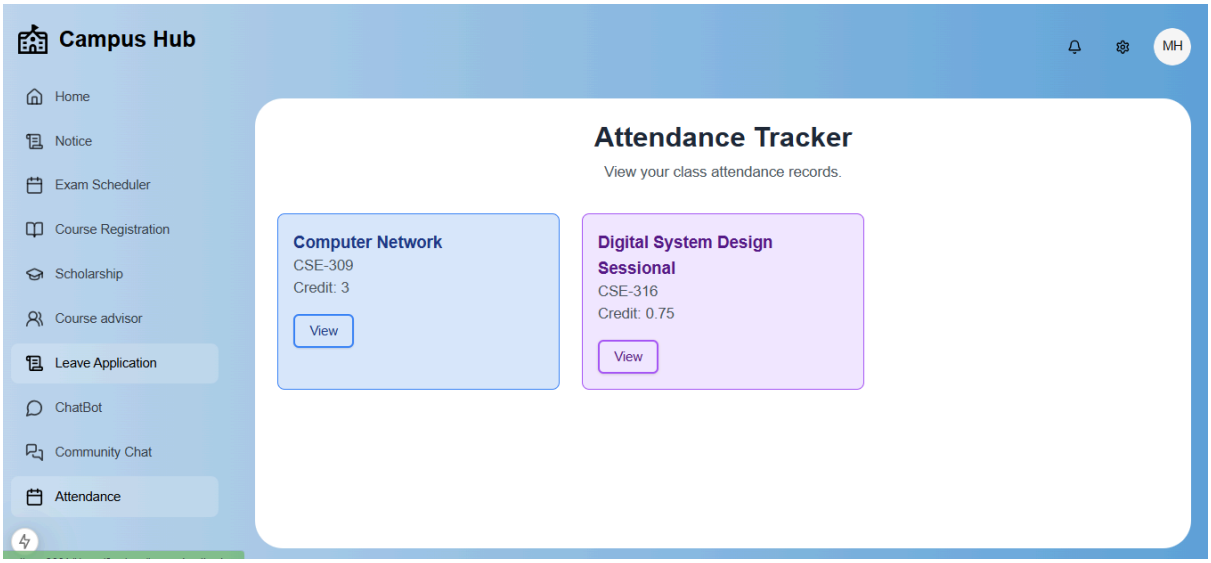


Fig 16 : Course list in Student view

Here student can see the attendance of particular course and approval or rejection for leave requests he applied for.

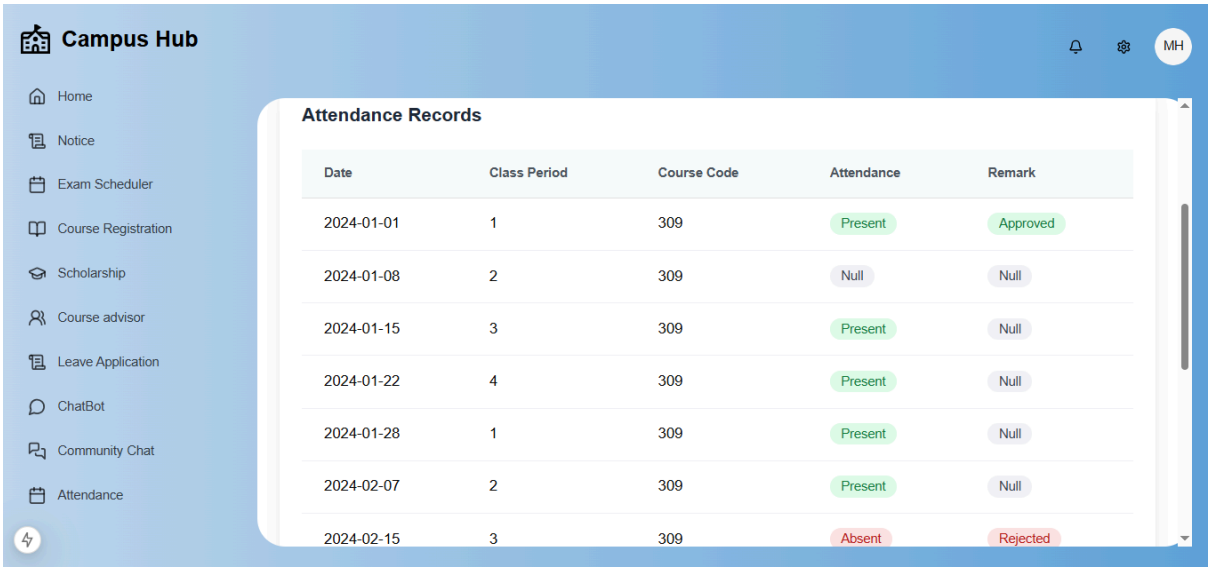


Fig 17 : Attendance in Student view

Students can ask any question to this Chatbot. Chatbot will respond accordingly.

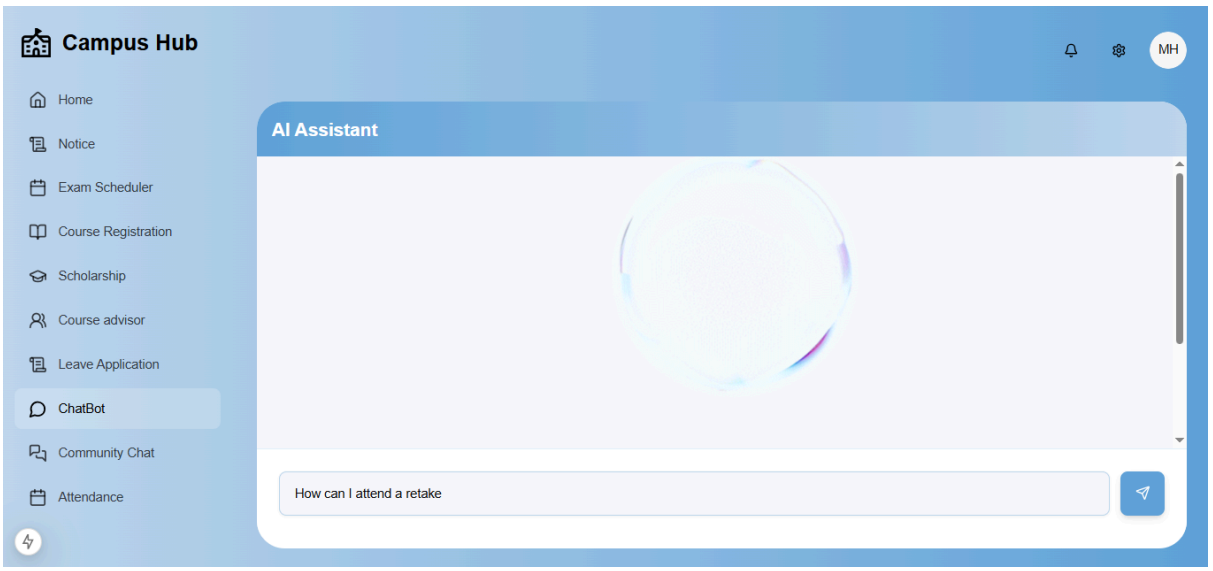


Fig 18 : Chatbot in student view

Here admin can train the chatbot.

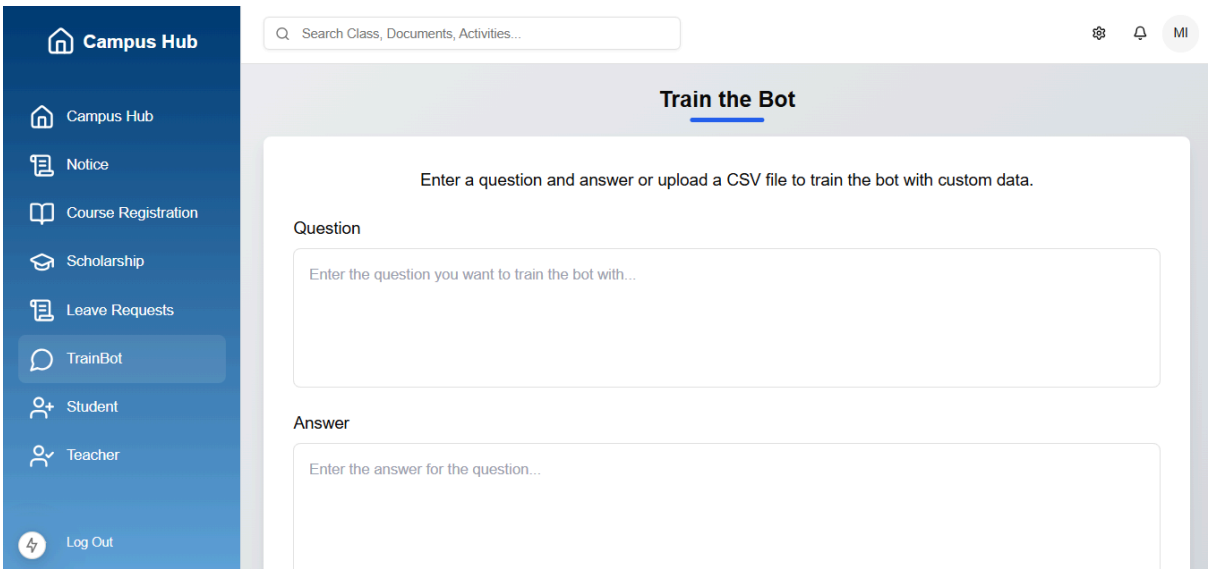


Fig 19: Train Chatbot

The Community Chat Feature is also available for the students where they can chat with each other as well as with the alumni.

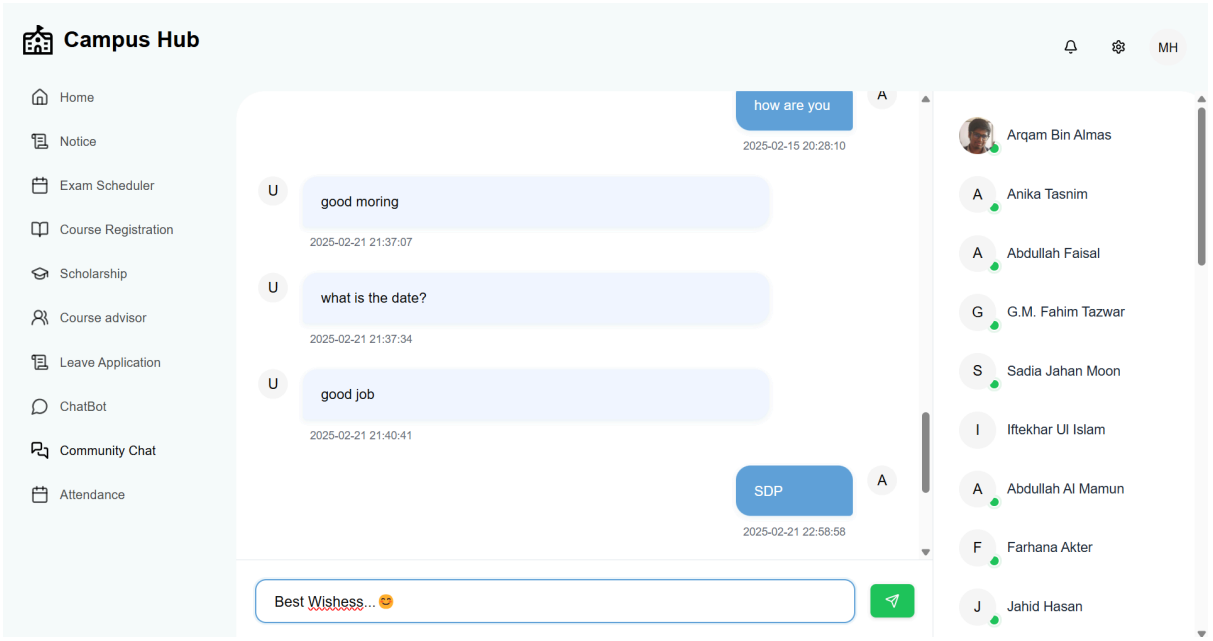


Fig 20: Community Chat

Students can also see any member's profile.

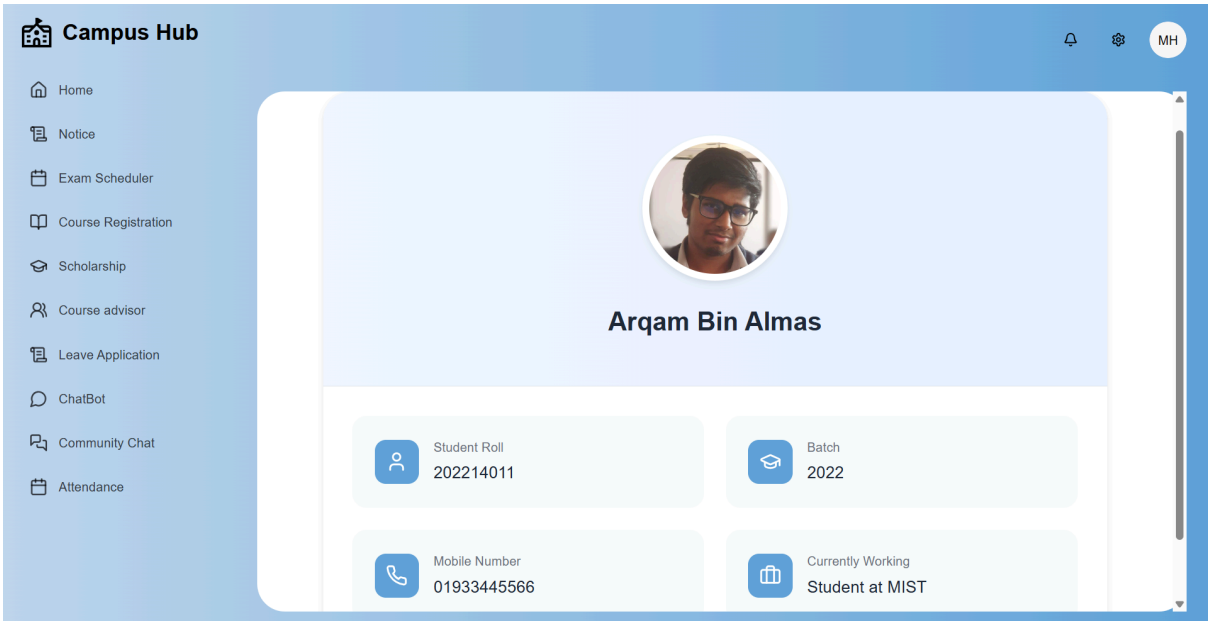


Fig 21: Member profile view from Community Chat

Admin can add new students.

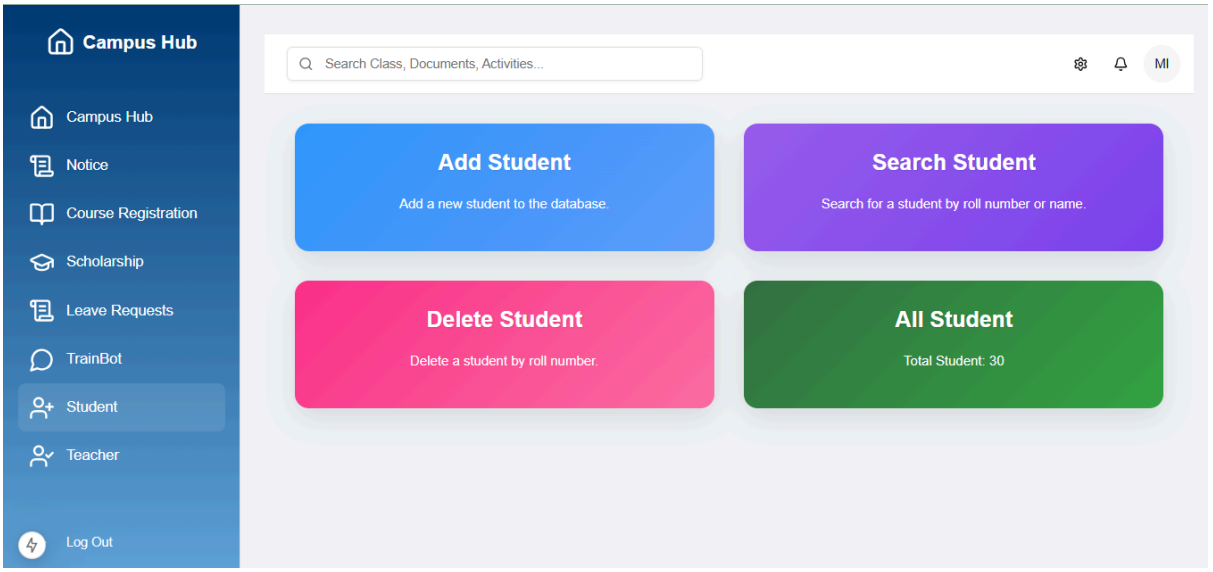


Fig 22 : Add new Student

Admin can add new Teacher.

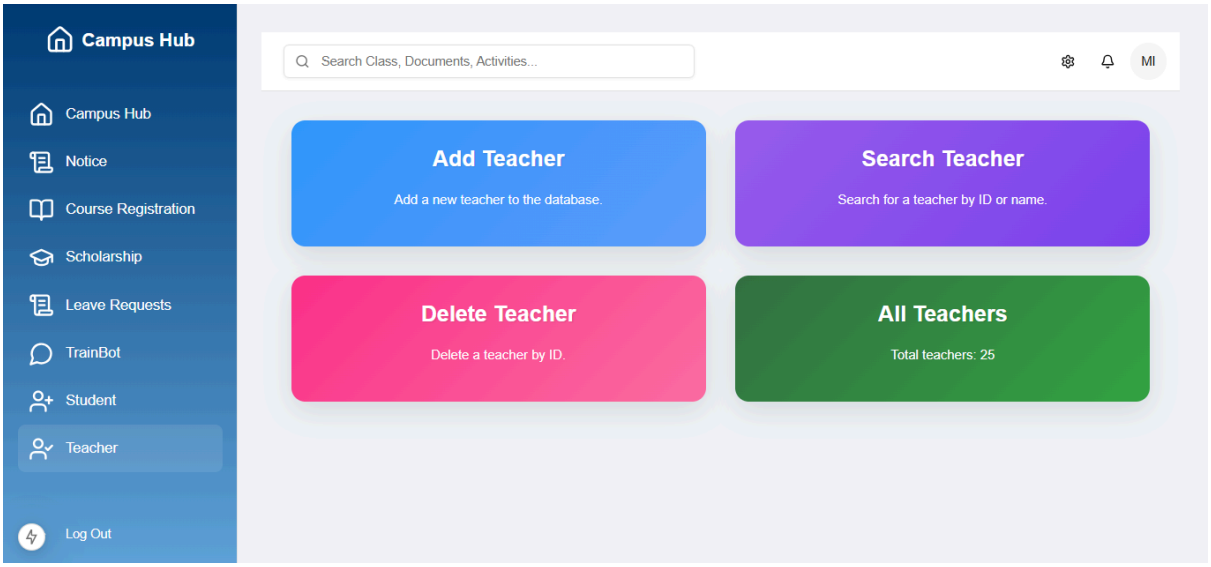


Fig 23: Add new teacher

In the Automated Course Advisor feature, student can see the list of curses where he/she can do better. The courses which result is less than GPA 3.25.

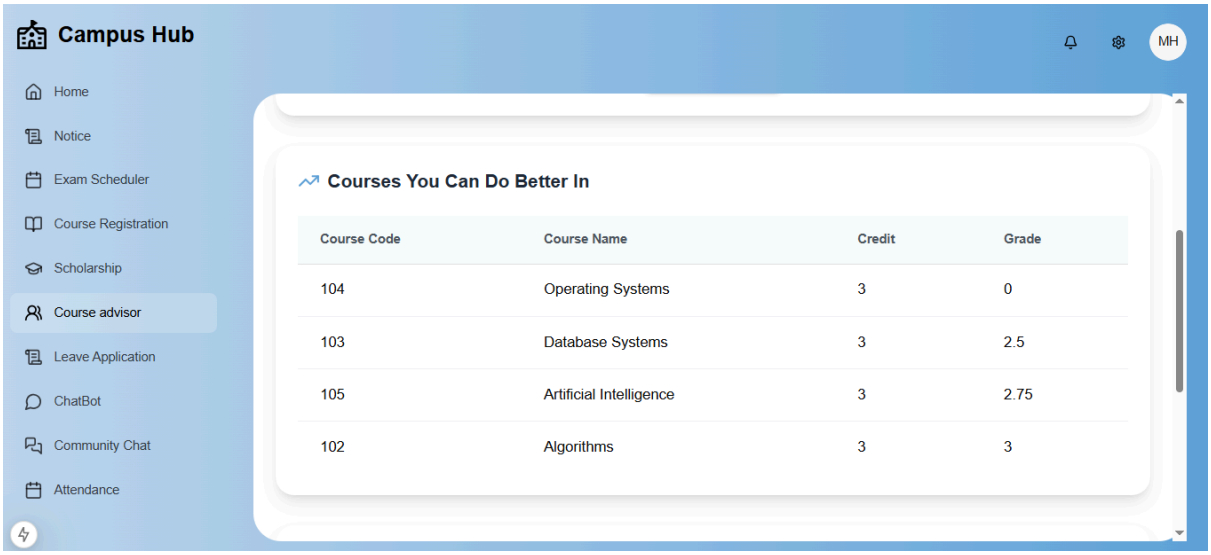


Fig 24 : Automated Course Advisor

Here student can also see the exam type he/she has to attend for those courses, like retake/supplement/improvement.

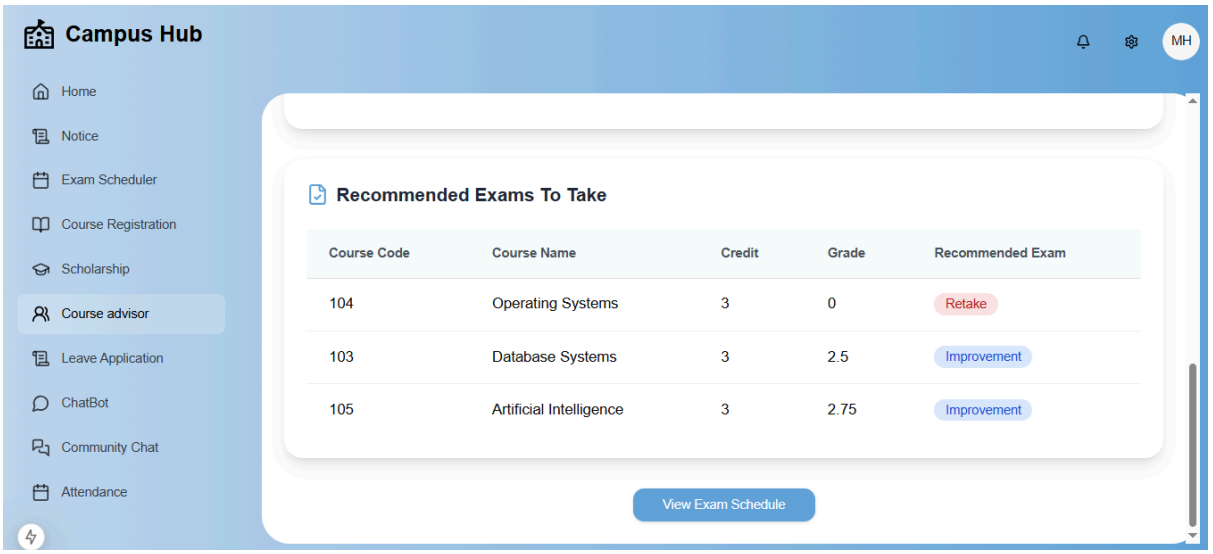


Fig 25 : Automated Course Advisor

13. Work Distribution

Name (ID)	Responsibilities
Sadia Jahan Moon (202114085)	• Automated Course Advisor • Leave application (Student) • Leave request approval (Admin) • Running courses (student) • Attendance (Student)
Mehedi Hasan Moon (202214048)	• Chatbot (Student) • Train Chatbot (Admin) • Attendance (Teacher) • Admin Profile • Teacher Profile
Maisha Nanjeeba (202214033)	• Exam Schedule (Student) • Exam Schedule (Teacher) • All User Dashboards • All User Login, Sign up
Md. Rubaet Kabir Zishan (202214054)	• Scholarship Application (Student) • Scholarship Verify and Forward (Teacher) • Scholarship Approval (Admin) • Course Registration (Student) • Course Approval (Admin)

Arqam Bin Almas (202214011)	● Notice (Admin,Student, Teacher) ● Community Chat (Student, Teacher) ● Student Profile
A S M Jobayer Hossain (202014072)	● Add new student (Admin) ● Add new teacher (Admin) ● User authentication and authorization

14. Limitations

- Not Capable of handling High User Load
- Difficulties integrating with current University systems
- Limited customization capability of Automated Course Advisor feature
- Users might have difficulties adapting
- Limited AI capabilities
- Can not take user feedback
- Mobile and IoS platforms not available

15. Future Scope

- Making the Services compatible with other platforms
- Integrate a user feedback section
- Expand AI chatbot capabilities
- Integrate with other platforms such as job portals
- Make it highly customizable so that any university can adapt to it

16. Conclusion

In conclusion, Campus Hub streamlines academic processes through automation and AI support. Our platform simplifies course advising, attendance tracking, leave applications, and scholarship management for students and faculty alike. While we continue to address current limitations, our plans include mobile applications and expanded AI capabilities.

17. References

University Grants Commission (UGC). (2022). *Annual report on higher education in Bangladesh.* Dhaka: UGC.

Bangladesh Bureau of Educational Information and Statistics (BANBEIS). (2022). *Educational statistics yearbook.* Dhaka: BANBEIS.

World Bank. (2021). *Education sector review: Leveraging technology for improved learning outcomes.* Washington, DC: World Bank.